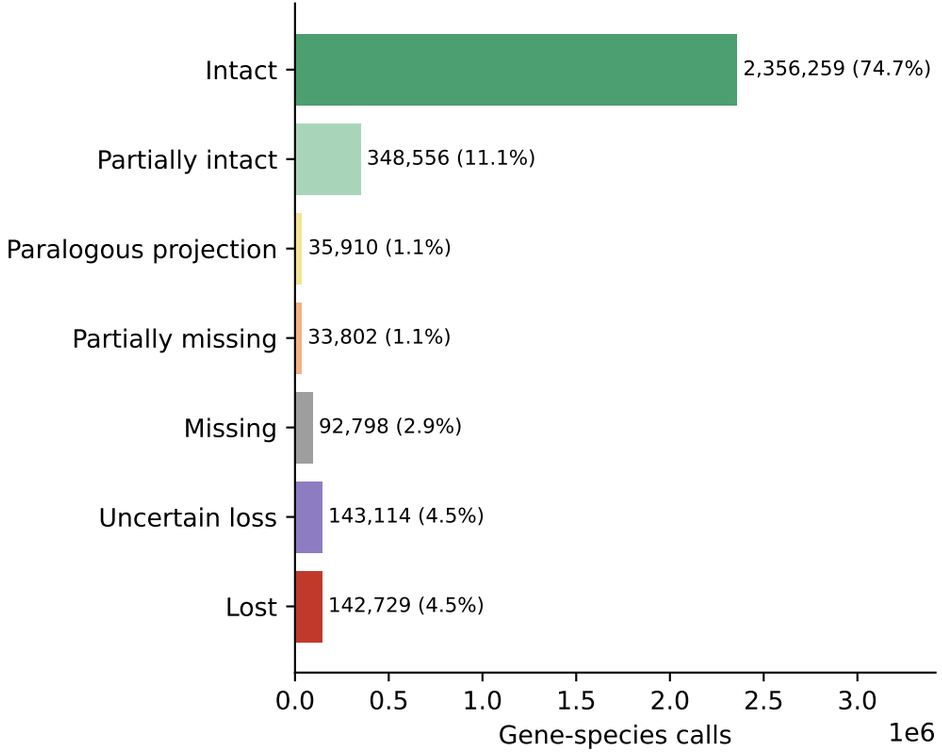
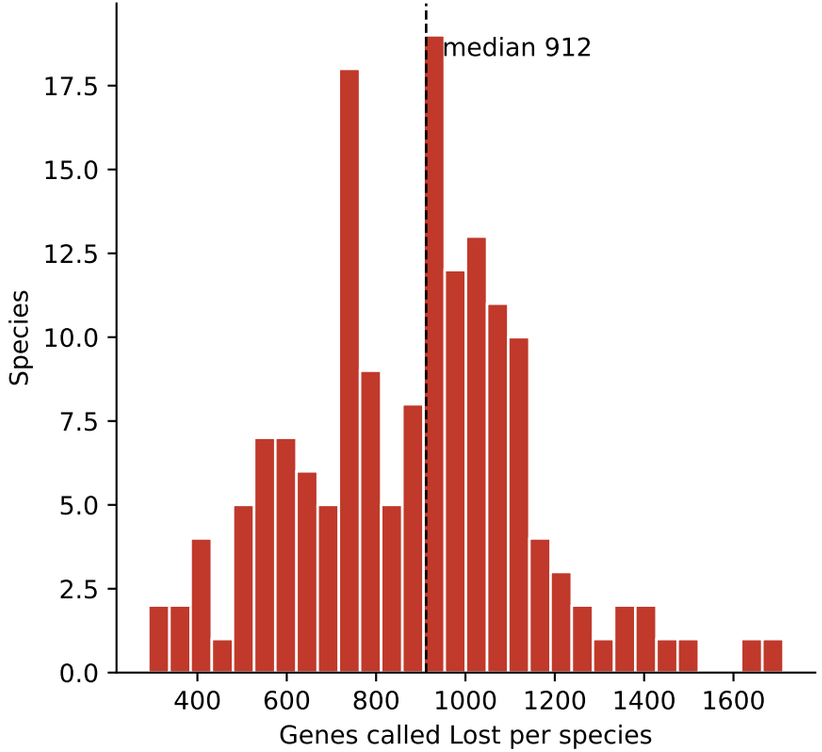
Gene status matrix: overview and QC

Status of all 3,153,168 calls

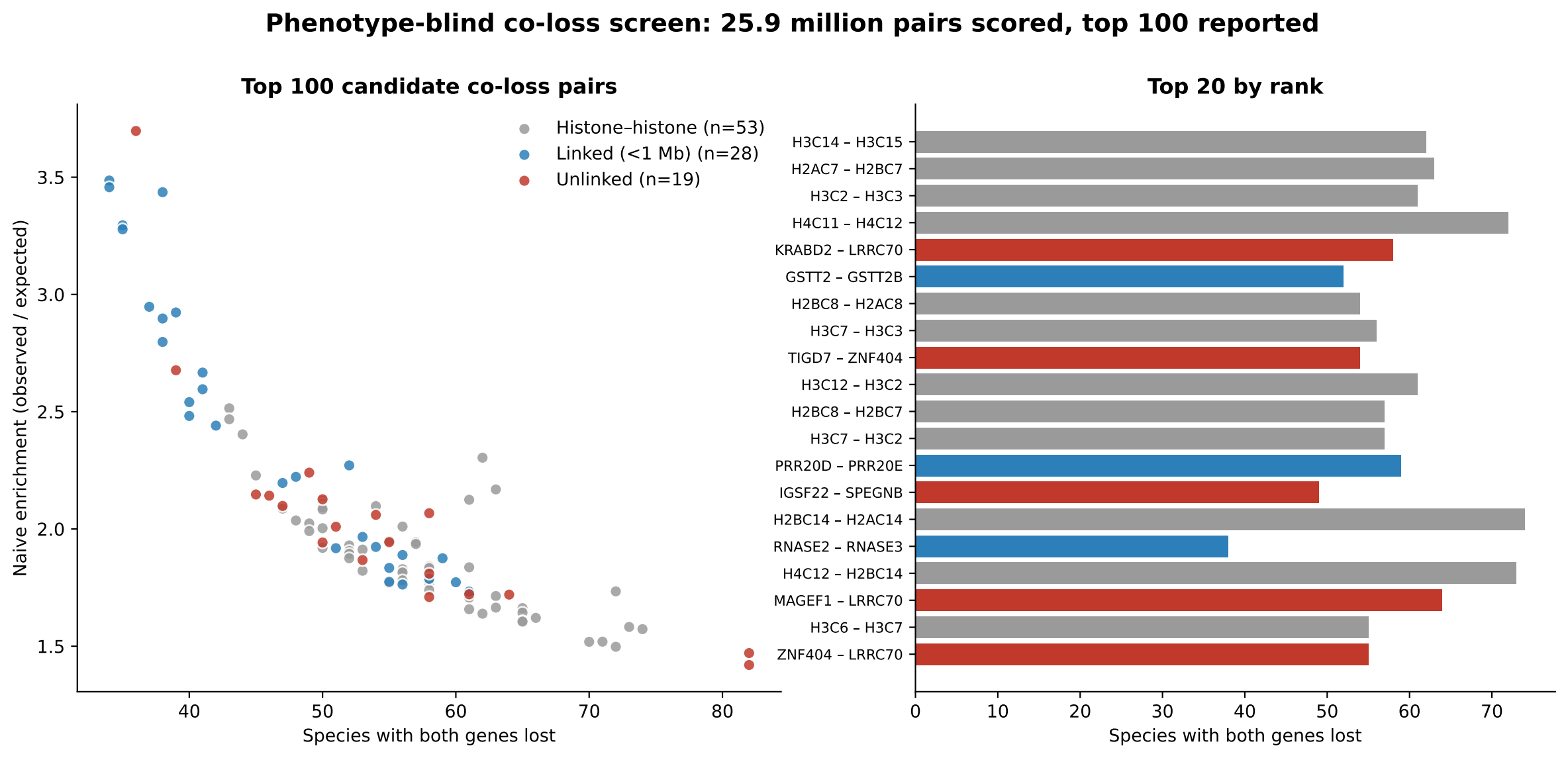


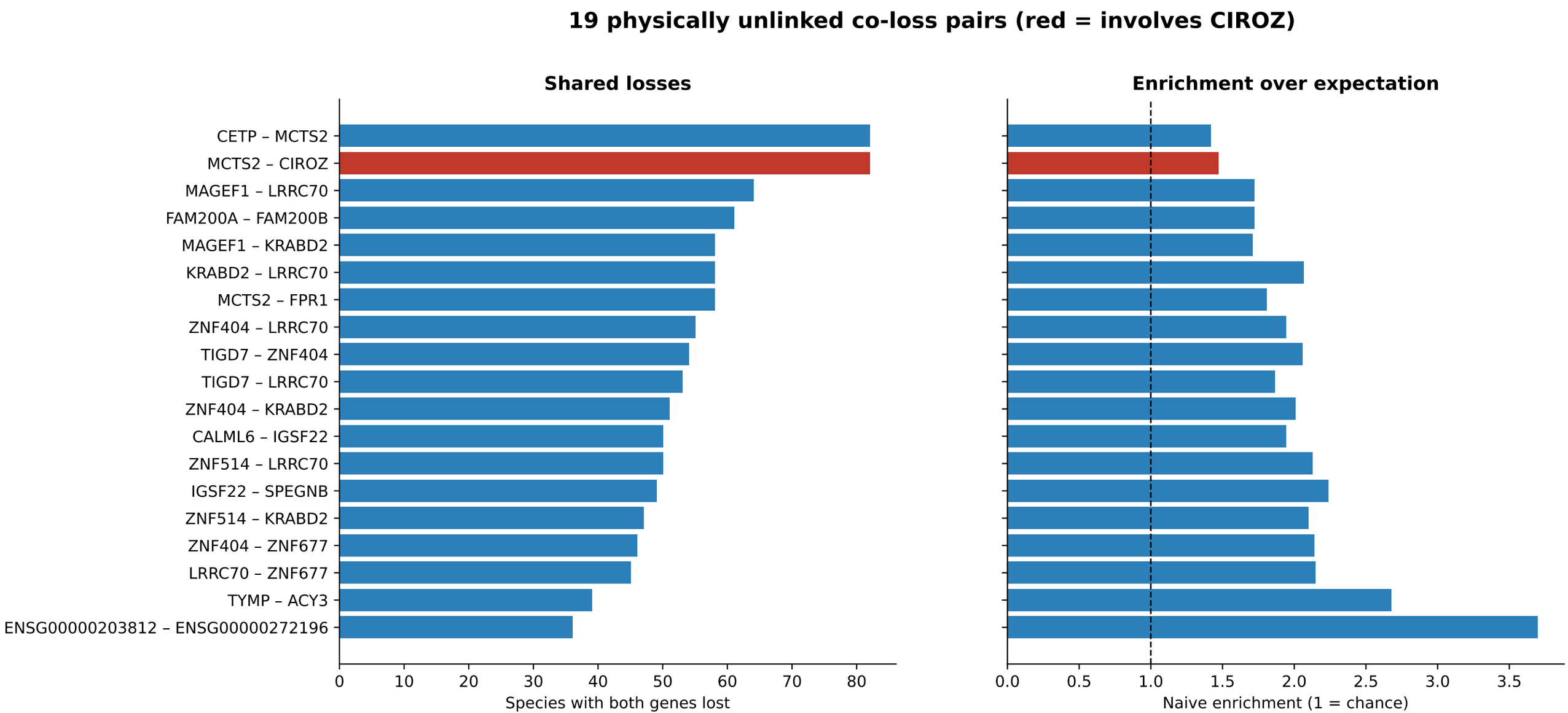
Loss burden per species

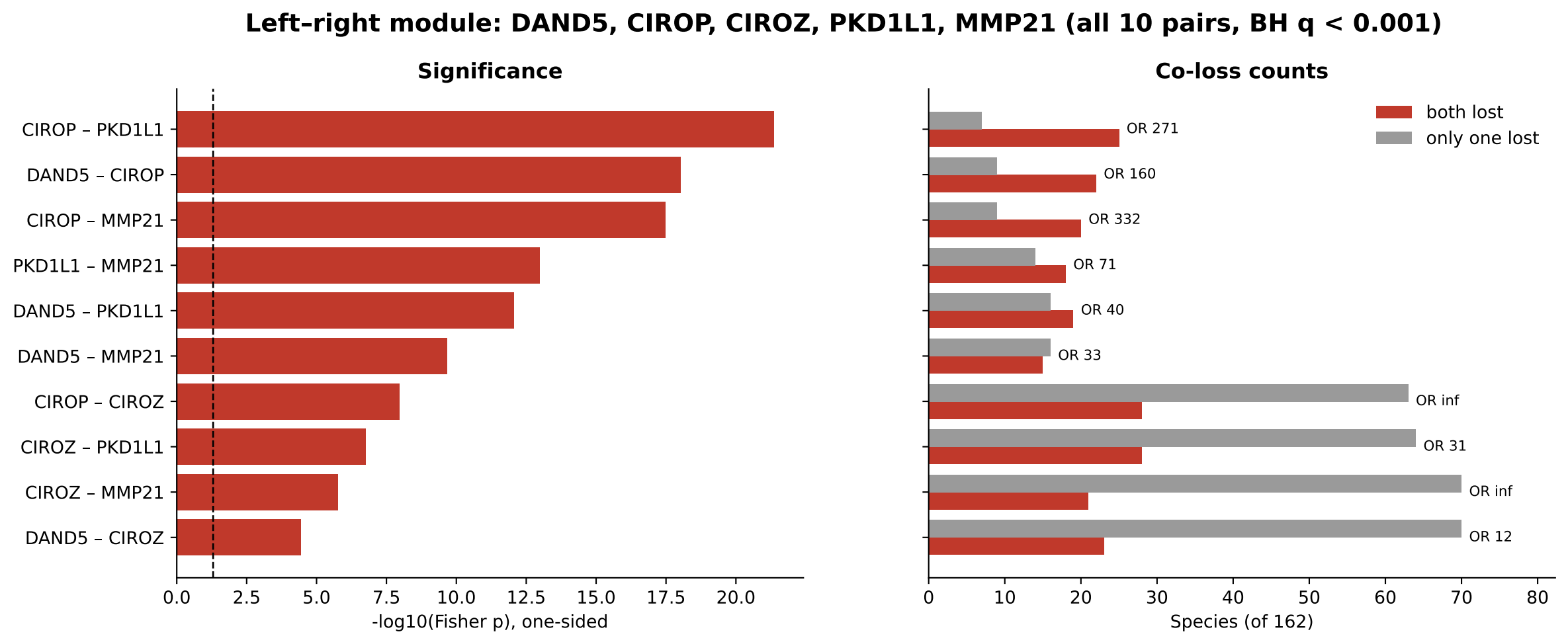


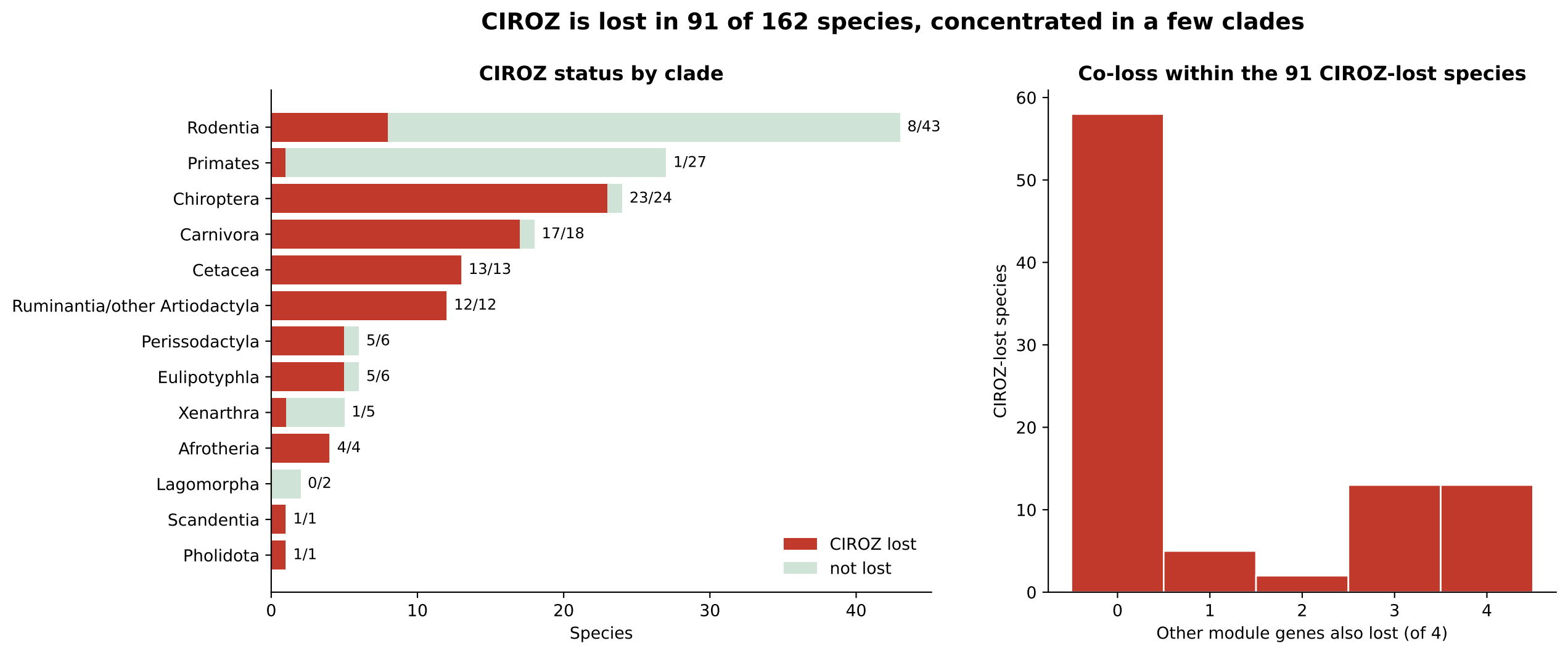
QC checks

Species	162
Genes	19,464
Matrix cells	3,153,168
Missing cells	0
Per-species source files	162 (SHA256 recorded)
Ancestry correction	not yet applied


Phenotype-blind co-loss screen: 25.9 million pairs scored, top 100 reported
Top 100 candidate co-loss pairs
Histone–histone (n=53)
Linked (<1 Mb) (n=28)
Unlinked (n=19)
Naive enrichment (observed / expected)
3.5
3.0
2.5
2.0
1.5
40
50
60
70
80
Species with both genes lost
Top 20 by rank
H3C14 – H3C15
H2AC7 – H2BC7
H3C2 – H3C3
H4C11 – H4C12
KRABD2 – LRRC70
GSTT2 – GSTT2B
H2BC8 – H2AC8
H3C7 – H3C3
TIGD7 – ZNF404
H3C12 – H3C2
H2BC8 – H2BC7
H3C7 – H3C2
PRR20D – PRR20E
IGSF22 – SPEGNB
H2BC14 – H2AC14
RNASE2 – RNASE3
H4C12 – H2BC14
MAGEF1 – LRRC70
H3C6 – H3C7
ZNF404 – LRRC70
0
10
20
30
40
50
60
70
Species with both genes lost


19 physically unlinked co-loss pairs (red = involves CIROZ)
Shared losses
Enrichment over expectation
CETP – MCTS2
MCTS2 – CIROZ
MAGEF1 – LRRC70
FAM200A – FAM200B
MAGEF1 – KRABD2
KRABD2 – LRRC70
MCTS2 – FPR1
ZNF404 – LRRC70
TIGD7 – ZNF404
TIGD7 – LRRC70
ZNF404 – KRABD2
CALML6 – IGSF22
ZNF514 – LRRC70
IGSF22 – SPEGNB
ZNF514 – KRABD2
ZNF404 – ZNF677
LRRC70 – ZNF677
TYMP – ACY3
ENSG00000203812 – ENSG00000272196
0
10
20
30
40
50
60
70
80
Species with both genes lost
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Naive enrichment (1 = chance)


Left–right module: DAND5, CIROP, CIROZ, PKD1L1, MMP21 (all 10 pairs, BH q < 0.001)
Significance
CIROP – PKD1L1
DAND5 – CIROP
CIROP – MMP21
PKD1L1 – MMP21
DAND5 – PKD1L1
DAND5 – MMP21
CIROP – CIROZ
CIROZ – PKD1L1
CIROZ – MMP21
DAND5 – CIROZ
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
-log10(Fisher p), one-sided
Co-loss counts
both lost
only one lost
OR 271
OR 160
OR 332
OR 71
OR 40
OR 33
OR inf
OR 31
OR inf
OR 12
0
10
20
30
40
50
60
70
80
Species (of 162)


CIROZ is lost in 91 of 162 species, concentrated in a few clades
CIROZ status by clade
Rodentia
Primates
Chiroptera
Carnivora
Cetacea
Ruminantia/other Artiodactyla
Perissodactyla
Eulipotyphla
Xenarthra
Afrotheria
Lagomorpha
Scandentia
Pholidota
8/43
1/27
23/24
17/18
13/13
12/12
5/6
5/6
1/5
4/4
0/2
1/1
1/1
CIROZ lost
not lost
0
10
20
30
40
Species
Co-loss within the 91 CIROZ-lost species
CIROZ-lost species
0
10
20
30
40
50
60
0
1
2
3
4
Other module genes also lost (of 4)